

NONABELIAN COMPOSITION FACTORS OF FINITE GROUPS WHOSE MAXIMAL SUBGROUPS ARE COMPLEMENTED

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ABSTRACT. Let G be a finite group such that every maximal subgroup M has an exact complement: $G = MK$ and $M \cap K = 1$ for some $K \leq G$. We prove that every nonabelian composition factor of G is isomorphic to

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \text{or} \quad \mathrm{PSL}_5(2).$$

Conversely, all three occur. This answers Kourovka Problem 18.68.

The main point is to avoid passing the hypothesis to sections, which is false. We first pass to a monolithic quotient with socle S^k . For any almost-simple coordinate group $S \leq X \leq \mathrm{Aut}(S)$, we manufacture a new primitive product action of that quotient. A theorem of Liebeck–Praeger–Saxl forces the coordinate action to have a core-free transitive subgroup. Maximal-factorization classifications exclude this for every simple S outside the displayed list. The only infinite factor-screen survivor, $S = \mathrm{PSp}_4(2^f)$, is eliminated by a 2-adic lifting argument using the fact that $O_4^+(2^f)$ meets every involution class. Fifteen named small almost-simple cases are certified by complete tables of marks; all infinite coverage comes from published exhaustive classifications.

1. INTRODUCTION

Say that a finite group G has property **CMP** if every maximal subgroup $M < G$ has a subgroup $K \leq G$ such that

$$(1) \quad G = MK, \quad M \cap K = 1.$$

The complement is not assumed normal. Kourovka Problem 18.68 asks for the nonabelian composition factors of a nonsoluble group with this property [5, Problem 18.68]. Levchuk and Likharev proved that the nonabelian simple groups having **CMP** are precisely $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_5(2)$ [7, Theorem 1]. The issue is that **CMP** does not pass to normal subgroups or sections. For example, D_8 has **CMP**, while its normal cyclic subgroup C_4 does not.

Our main result is the expected section-safe extension of the simple-group classification.

Theorem 1.1. *Let G be a finite group with **CMP**. Every nonabelian composition factor of G is isomorphic to one of*

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \mathrm{PSL}_5(2).$$

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Each of the three possibilities occurs.

The proof uses the classification of finite simple groups (CFSG), the maximal-factorization classifications cited in [Section 4](#), and Corollary 3 of Liebeck–Praeger–Saxl [9]. The finite calculation in [Section 7](#) concerns only fifteen explicitly named almost-simple groups. No bounded computation is used to infer an infinite family statement.

2. FROM A COMPOSITION FACTOR TO A MONOLITHIC QUOTIENT

We begin with the inheritance statements that are actually valid.

Lemma 2.1 (quotient closure). *If G has CMP and $N \trianglelefteq G$, then G/N has CMP.*

Proof. Let $\overline{M}$ be maximal in G/N , and let M be its full preimage. Choose $K \leq G$ satisfying (1). Then $G/N = \overline{M}(KN/N)$. If $kN \in \overline{M}$ with $k \in K$, then $k \in M$, and hence $k = 1$. Thus KN/N is an exact complement to $\overline{M}$. $\square$

Lemma 2.2 (regular-action reformulation). *A subgroup $K \leq G$ satisfies (1) if and only if it is regular on the right cosets of M . If M is maximal and core-free, this is a faithful primitive action.*

Proof. Transitivity is equivalent to $G = MK$, and the stabilizer in K of the coset M is $M \cap K$. $\square$

Proposition 2.3 (monolithic reduction). *Let S be a nonabelian composition factor of a finite CMP-group G . There is a quotient L of G such that*

- (1) L has CMP;
- (2) $\text{Soc}(L) = N \cong S^k$ for some $k \geq 1$, and N is the unique minimal normal subgroup of L ;
- (3) $C_L(N) = 1$.

Proof. Choose a chief factor $A/B \cong S^k$ of G , and define

$$C/B = C_{G/B}(A/B), \quad L = G/C, \quad N = AC/C.$$

Since A/B is centerless, $A \cap C = B$, and hence $N \cong A/B$. Chiefness makes N a minimal normal subgroup of L . The conjugation action of L on N is faithful by the definition of C , so $C_L(N) = 1$. If R were another minimal normal subgroup, then $[R, N] \leq R \cap N = 1$, forcing $R \leq C_L(N) = 1$. Thus N is unique. Finally, L has CMP by [Lemma 2.1](#). $\square$

We shall use a stronger fact than the existence of one primitive action: the same abstract group L has many primitive actions, one for each suitable action of its coordinate group.

3. THE PRODUCT-ACTION LIFTING THEOREM

Write

$$N = S_1 \times \cdots \times S_k \cong S^k.$$

The action of L on the factors is transitive, since the product of the factors in any orbit is normal in L . Set

$$(2) \quad X = N_L(S_1)/C_L(S_1),$$

identifying S_1 with its inner automorphism group. Then $S \leq X \leq \text{Aut}(S)$. The standard wreath embedding gives

$$(3) \quad L \hookrightarrow X \wr P \leq X \wr \text{Sym}(k),$$

where P is transitive, N is the base socle S^k , and the group induced on a coordinate is X . This embedding is faithful because $C_L(N) = 1$.

Let $H < X$ be core-free and maximal. Put $\Delta = X/H$ and $V = H \cap S$, and assume $V \neq 1$. Since S is normal and is not contained in H , we have $X = SH$; in particular, S is transitive on Δ . Via (3), let L act on Δ^k in product action.

Lemma 3.1 (primitivity). *The action of L on Δ^k is faithful and primitive.*

Proof. Let $\omega = (H, \dots, H)$, $M = L_\omega$, and $N = S^k$. The group N is transitive, so $L = MN$ and $M \cap N = V^k$.

Suppose $M \leq J \leq L$, and put $D = J \cap N$. Then $J = MD$, $D \trianglelefteq M$, and $V^k \leq D$. The image of M on the coordinates is transitive: multiply an element of L by a suitable base element to make it fix ω without changing its top permutation. Moreover, the subgroup of M stabilizing coordinate i induces H there. The nontrivial inclusion follows by taking an element inducing a prescribed $h \in H$ and correcting all other coordinates with elements of N .

Let D_i be the projection of D to $S_i \cong S$. It contains V and is normalized by H . Maximality of H implies either $D_i = V$ or $HD_i = X$. In the second case

$$D_i = (H \cap S)D_i = (HD_i) \cap S = X \cap S = S.$$

Coordinate transitivity gives the same alternative for every i . If all projections are V , then $D = V^k$ and $J = M$. Otherwise D is subdirect in S^k . By Scott's lemma it is a direct product of full diagonal strips. Since $V^k \leq D$ and $V \neq 1$, every strip has length one, so $D = N$ and $J = L$. Thus M is maximal. Faithfulness follows from core-freeness of H and (3). $\square$

Call (X, Δ) *factor-free* if it has no core-free transitive subgroup; equivalently, there is no core-free $C < X$ with $X = HC$.

Proposition 3.2 (coordinate obstruction). *If $(X, X/H)$ is factor-free, then L does not have CMP.*

Proof. For $k = 1$, a regular subgroup on X/H would itself be core-free and transitive. Let $k \geq 2$. The action in Lemma 3.1 is a primitive product-action with almost-simple component $(X, X/H)$: indeed, its socle stabilizer is V^k , which is nontrivial and is not subdirect in S^k . Corollary 3(iv) of [9] says that if such a group has a regular subgroup, then its component action has a core-free transitive subgroup. Hence no regular subgroup exists. But if L had CMP, its maximal point stabilizer would have a complement, regular by Lemma 2.2, a contradiction. $\square$

This is the step that removes the unbounded exponent k : it converts the entire product-action problem into one almost-simple factorization screen.

4. FACTOR SCREENS AND PRIME LIFTING

We first state the mechanism for applying maximal-factorization tables.

Lemma 4.1 (factor screen). *Let $1 < V < S$, and assume that the S -conjugacy class of V is invariant under X . Choose a maximal subgroup $H < X$ containing $N_X(V)$. Then H is core-free and $X = HS$. If X/H is not factor-free, there are $S \leq Y \leq X$ and a nontrivial maximal factorization $Y = AB$ such that, after interchanging the factors,*

$$V \leq A \cap S.$$

Proof. Class invariance gives $N_X(V)S = X$, so no proper overgroup of $N_X(V)$ contains S . Hence H is core-free and $HS = X$.

Suppose $C < X$ is core-free and $X = HC$. Put $Y = CS$. Then $Y = (H \cap Y)C$, and both factors supplement S in Y . Consequently all their proper maximal overgroups are core-free. Enlarging them gives a maximal factorization $Y = AB$, with $V \leq H \cap S \leq A \cap S$. $\square$

The remaining tool is the prime-divisibility analogue of the obstruction used in related subgroup-product problems.

Definition 4.2. An action of Z on Λ is *p-elusive* if $p \mid |\Lambda|$ and every element of order p in Z fixes a point.

Lemma 4.3 (wreath lifting). *If X is p-elusive on Δ , then $X \wr \text{Sym}(k)$ is p-elusive on Δ^k . In particular, no subgroup of this wreath product is regular on Δ^k .*

Proof. Let $w = (x_1, \dots, x_k)\sigma$ have order p . On a fixed coordinate of σ , the corresponding x_i has order 1 or p , and hence fixes a point. On a p -cycle of σ , the cycle product of the x_i is 1, since $w^p = 1$; the fixed-tuple equations can therefore be solved successively around the cycle. Thus w fixes a tuple.

If a subgroup were regular, Cauchy's theorem would give it a nonidentity element of order p , because its order is $|\Delta|^k$. That element would fix a point, a contradiction. $\square$

Lemma 4.4 (socle-valuation lifting). *Suppose S is p -elusive on $\Delta = X/H$. Let*

$$a = v_p(|\Delta|), \quad o = v_p(|X/S|).$$

If $a > o$, then no subgroup $L \leq X \wr \text{Sym}(k)$ containing S^k is regular on Δ^k .

Proof. Suppose R is regular and put $Q = R \cap S^k$. Since $R/Q \hookrightarrow L/S^k$,

$$v_p(|R/Q|) \leq ko + v_p(k!).$$

As $v_p(|R|) = ka$,

$$v_p(|Q|) \geq k(a - o) - v_p(k!) > 0,$$

because $a - o \geq 1$ and $v_p(k!) < k$. An element of order p in Q fixes a tuple coordinatewise, contrary to regularity. $\square$

5. THE COORDINATE CLASSIFICATION

Proposition 5.1. *Let S be a nonabelian finite simple group not isomorphic to $\text{PSL}_2(7)$, $\text{PSL}_2(11)$, or $\text{PSL}_5(2)$, and let $S \leq X \leq \text{Aut}(S)$. There is a core-free maximal subgroup $H < X$, with $H \cap S \neq 1$, for which one of the following holds:*

- (1) X/H is factor-free;
- (2) X is p -elusive on X/H for some prime p ;
- (3) $S = \text{PSp}_4(2^f)$, $f \geq 2$, and the hypotheses of [Lemma 4.4](#) hold with $p = 2$.

Proof.

5.1. Alternating groups. For $n \neq 6$, the coordinate group is A_n or S_n . The following actions are p -elusive:

Socle	action and degree	prime
A_5	2-subsets, 10	2
A_6 , $X = A_6$	natural, 6	2
A_6 , $X = S_6$	unordered 3 + 3 partitions, 10	2
A_8	2-subsets, 28	2
A_9	2-subsets, 36	2
A_{10}	4-subsets, 210	2
A_{11}	3-subsets, 165	3
A_{12}	2-subsets, 66	2

The assertion is immediate from cycle supports. An involution fixes the support of one 2-cycle. In degree 10, every involution also has an invariant 4-set: use two 2-cycles, or one 2-cycle and two fixed points. An element of order 3 in degree 11 fixes the support of one 3-cycle. For the S_6 partition action, the cycle types $2, 2^2, 2^3$ respectively preserve a 3-set or interchange it with its complement.

The three outer extensions of A_6 not equal to S_6 , and the two groups A_7, S_7 , have factor-free maximal subgroups certified in [Section 7](#). Their orders and indices are

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Finally let $n \geq 13$, and take the stabilizer of a 6-subset. It is maximal because $6 < n/2$. The alternating-group part of the LPS maximal factorization theorem [[8](#), Theorem D and Remark 2] says that an intransitive factor in a nontrivial maximal factorization stabilizes a j -subset with $j \leq 5$, apart from the listed cases of degrees 6, 8, 10. Thus the 6-set action is factor-free.

5.2. Classical groups. We use Theorem 2.15 and Appendix Tables A.1–A.7 of Xia–Li [[11](#)], a restatement of the LPS classification of nontrivial maximal factorizations with classical socle, together with the two corrected $\text{P}\Omega_8^+(4)$ and $\text{P}\Omega_8^+(16)$ rows of Gill–Giudici–Spiga [[3](#), Section 2]. In the table below, V is invariant up to S -conjugacy under X . Applying [Lemma 4.1](#), inspection of the indicated factorization table shows that no factor intersection contains V .

Socle and conditions	screen subgroup V	table check
$\text{PSL}_2(q)$, excluding the allowed and alternating cases	split-torus normalizer	not in either torus type or P_1 ; exceptional factors are checked below
$\text{PSL}_3(q)$, $q \neq 4$, excluding $q = 2$	split-torus normalizer	not in a point or hyperplane parabolic or in the extension-field factor
$\text{PSL}_3(4)$, every coordinate group X	a pinned maximal subgroup of X	complete Table-of-Marks certificate in Section 7
$\text{PSL}_n(q)$, even $n \geq 4$	$P_{n/2}$	absent from A.1
$\text{PSL}_n(q)$, odd $n \geq 5$, no graph	P_2	absent from A.1 except $(n, q) = (5, 2)$
$\text{PSL}_n(q)$, odd $n \geq 5$, graph present	$P_{2, n-2}$	only parabolic overgroups P_2, P_{n-2} ; absent except $(5, 2)$
$\text{PSp}_{2m}(q)$, $m \geq 3$	P_2	absent from A.2
$\text{PSp}_4(q)$, q odd	nonsingular $2 + 2$ decomposition stabilizer	absent from A.2
$\text{PSU}_3(q)$	N_1	absent from the $n = 3$ rows of A.3

Socle and conditions	screen subgroup V	table check
$\text{PSU}_4(q), q \geq 3$	full nonsingular $2 + 2$ decomposition stabilizer	absent from A.3
$\text{PSU}_n(q), n \geq 5$	N_2	absent from A.3
$\Omega_{2m+1}(q), m \geq 3, q$ odd	P_2	absent from A.4
$\text{P}\Omega_{2m}^-(q), m \geq 4$	P_2	absent from A.5
$\text{P}\Omega_{2m}^+(q), m \geq 5$	P_2	absent from A.6
$\text{P}\Omega_8^+(q)$, triality allowed	central-node P_2	fixed by triality and absent from A.7

The two corrected orthogonal factorizations have socle intersections N_2^- and a triality image of $\Omega_8^-(q^{1/2})$, for $q = 4, 16$. Neither contains the central-node maximal parabolic P_2 , so they do not alter the last row of the screen.

Here a standard maximal parabolic can only be contained in a proper factor intersection if it is that intersection. A subgroup containing $P_{2,n-2}$ contains a Borel and is therefore parabolic; its only proper overgroups are P_2 and P_{n-2} . These observations justify the containment, rather than mere equality, assertions in the table. The displayed N_i and full $2 + 2$ decomposition stabilizers are standard maximal geometric subgroups for the stated parameters [6, Chapters 3–4]; the exceptional $\text{PSU}_4(2)$ parameter is routed through the low-rank isomorphism below.

For the first row, the split-torus normalizer has order $2(q-1)/(2, q-1)$ and interchanges the two fixed points of its torus, so it is not contained in P_1 . Nor can it lie in the nonsplit-torus normalizer: its order would force $q-1 \mid q+1$, apart from nonsimple parameters. The other A.1 factors occur only for

$$q = 7, 11, 16, 19, 23, 29, 59.$$

The values 7, 11 are allowed. At $q = 19, 29, 59, 23$, divisibility by the torus-normalizer order excludes A_5 or S_4 . At $q = 16$, a subgroup of order 30 lies neither in D_{34} nor in A_5 , the latter because it would have index 2. This closes the rank-one family.

For $S = \text{PSL}_3(q)$, let $d = (3, q-1)$ and let V normalize a maximally split torus. Its class is invariant under diagonal, field, and graph automorphisms. The monomial action shows that V fixes neither a point nor a hyperplane, and

$$|V| = \frac{6(q-1)^2}{d}.$$

The extension-field factor in row 1 of Table A.1 has intersection with S of order $3(q^2 + q + 1)/d$. For $q = 3$, divisibility excludes containment; for $q \geq 5$, the former order is larger. The only remaining table row is the

exceptional $q = 4$ row, and all ten almost-simple groups with that socle are handled by the finite certificate. The parameter $q = 2$ gives the allowed group $\mathrm{PSL}_3(2) \cong \mathrm{PSL}_2(7)$.

It remains to treat $S = \mathrm{PSp}_4(q)$ with $q = 2^f$, $f \geq 2$. Let $U = \Omega_4^+(q)$ and $H = N_X(U)$. Case (10.1) of [10, Section 10] gives that H is core-free maximal and $H \cap S = O_4^+(q)$. Thus $X = HS$, and hence

$$(4) \quad |X : H| = |S : O_4^+(q)| = \frac{q^2(q^2 + 1)}{2}, \quad v_2(|X : H|) = 2f - 1.$$

Corollary 2.3 of [10] shows that $O_4^+(q)$ meets every involution class of S , so S is 2-elusive on X/H . Moreover, the standard outer-automorphism description [11, Table 2.1] gives $|X/S| \mid 2f$, and hence

$$v_2(|X/S|) \leq 1 + v_2(f) < 2f - 1.$$

The last strict inequality follows, for example, from $v_2(f) \leq f - 1$ and $f \geq 2$. Now Lemma 4.4 applies. This is the sole infinite classical row for which the factor screen itself does not suffice.

The standard low-rank isomorphisms

$$\begin{aligned} \mathrm{PSL}_2(4) &\cong \mathrm{PSL}_2(5) \cong A_5, & \mathrm{PSL}_2(9) &\cong A_6, \\ \mathrm{PSL}_3(2) &\cong \mathrm{PSL}_2(7), & \mathrm{PSL}_4(2) &\cong A_8, \\ \mathrm{PSU}_4(2) &\cong \mathrm{PSp}_4(3). \end{aligned}$$

and the $B_2 = C_2$, $D_3 = A_3$, ${}^2D_3 = {}^2A_3$ identifications route every overlap to a row already considered.

5.3. Exceptional groups of Lie type. Hering–Liebeck–Saxl determine all factorizations of exceptional simple groups and their automorphism groups [4, Theorems 1–2]. The only socles admitting a core-free factorization are

$$G_2(4), \quad G_2(3^f), \quad F_4(2^f),$$

together with one extra outer factorization for $G_2(4).2$. None of the listed factors is parabolic. For every exceptional socle outside these three families, take V to be a Sylow subgroup in the defining characteristic. Its S -conjugacy class is invariant under every automorphism, and the Hering–Liebeck–Saxl theorem and Lemma 4.1 make the resulting action factor-free.

It remains to screen the three factorable families. If no graph automorphism is present, choose a maximal parabolic. In the graph-fused cases choose a Borel $P_1 \cap P_2$ in $G_2(3^f)$, or $P_1 \cap P_4$ in $F_4(2^f)$. Their respective p -parts are q^6 and at least q^{24} . The p -parts of the displayed Hering–Liebeck–Saxl factors are at most q^3 and q^{16} , so no factor contains the chosen screen subgroup. The small isomorphisms $G_2(2)' \cong \mathrm{PSU}_3(3)$ and ${}^2G_2(3)' \cong \mathrm{PSL}_2(8)$ have already been covered.

5.4. Sporadic groups. Giudici classifies all factorizations with sporadic socle [2, Theorems 1.1–1.3]. Comparing those tables with the Atlas maximal-subgroup lists [1] gives the following factor-free maximal subgroups. A row for an outer extension also excludes the factorizations lifted from the simple socle.

X	H	X	H
M_{11}	S_5	M_{12}	$2^3:S_4$
$M_{12}.2$	$3^{1+2}:D_8$	$M_{22}.2$	$A_6.2^2$
M_{23}	A_8	J_2	$3.A_6.2_2$
$J_2.2$	$3.A_6.2^2$	HS	$L_3(4).2_1$
$HS.2$	$L_3(4).2^2$	He	$2^2.L_3(4).S_3$
$He.2$	$2^2.L_3(4).D_{12}$	Ru	$2^6:U_3(3):2$
Suz	$J_2.2$	$Suz.2$	$J_2.2 \times 2$
Fi_{22}	$O_7(3)$	$Fi_{22}.2$	$G_2(3).2$
Co_1	$2^{11}:M_{24}$		

The simple group M_{22} , and every sporadic almost-simple group not needed in this display, has no nontrivial core-free factorization. For such a group, apply Lemma 4.1 to a Sylow subgroup of the socle; its conjugacy class is automorphism-invariant and its maximal overgroup has nontrivial socle intersection. The rows available in TomLib are independently checked in Section 7.

The sole factor-cover survivor is M_{24} . Let it act on the 2-subsets of its natural 24-point set. The point stabilizer $M_{22}.2$ is maximal, the degree is 276, and every involution fixes the support of one of its 2-cycles. Thus the action is 2-elusive and Lemma 4.3 applies.

This completes the CFSG audit. $\square$

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Let S be a nonabelian composition factor of G . Apply Proposition 2.3 to obtain a CMP-quotient L with unique self-centralizing socle S^k , and define its coordinate group X by (2).

Suppose S is not one of the three groups in the conclusion. Choose the maximal subgroup $H < X$ supplied by Proposition 5.1. By Lemma 3.1, the same group L has a faithful primitive action on $(X/H)^k$. If the action X/H is factor-free, Proposition 3.2 contradicts $\text{CMP}(L)$. If it is p -elusive, Lemma 4.3 shows that the primitive action has no regular subgroup, again contradicting $\text{CMP}(L)$. In the final symplectic case, the same contradiction follows from Lemma 4.4. Therefore S is one of the three displayed groups.

Conversely, Levchuk–Likharev prove that each of the three simple groups has CMP [7, Theorem 1]; taking the group itself shows that each occurs as a composition factor. $\square$

7. FINITE TABLE-OF-MARKS CERTIFICATE

The proof uses finite certificates for five small alternating coordinate groups and for the ten almost-simple coordinate groups with socle $\text{PSL}_3(4)$. Nine sporadic rows are also cross-checked.

For a complete table of marks T , let h, c, i represent conjugacy classes of subgroups H, C, I . The command `IntersectionsTom(T,h,c)[i]` is positive exactly when suitable conjugates of H, C have intersection in class i . Such a pair factors X precisely when

$$|H||C| = |X||I|.$$

A subgroup C is core-free in an almost-simple X precisely when its class does not contain the socle class. The producer checks every core-free class c and every intersection class i , verifies $|H \cap S| = |H||S|/|X| > 1$, and fails on missing data.

The proof rows are:

X	$A_6.2_2$	$A_6.2_3$	$A_6.2^2$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Since $\text{Out}(\text{PSL}_3(4)) \cong D_{12}$, the following ten TomLib names exhaust, up to isomorphism, the groups between the socle and its full automorphism group:

X	$ H $	$ X:H $	X	$ H $	$ X:H $
$L_3(4)$	960	21	$L_3(4).2_1$	384	105
$L_3(4).2_2$	1920	21	$L_3(4).2_3$	720	56
$L_3(4).3$	216	280	$L_3(4).2^2$	768	105
$L_3(4).3.2_2$	432	280	$L_3(4).3.2_3$	1152	105
$L_3(4).6$	1152	105	$L_3(4).D_{12}$	2304	105.

Every row has zero core-free factor classes. The producer and independent checker are, respectively,

```
gap -q gap/generate-factor-free-scan.g
python3 tests/check-factor-free-scan.py
```

in the accompanying repository. The environment is GAP 4.15.1 with TomLib 1.2.11. The generated TSV has SHA-256

82bcf695617014f0124839c5a01983a6c8904fc5bc92e0893c9c2601c43bd3a0

and records all twenty-four proof and cross-check rows. This calculation is an exact finite verification relative to the complete named tables, not an extrapolation from a library scan.

8. DEPENDENCY AND SCOPE STATEMENT

The universal conclusion depends on CFSG and on the exhaustive published factorization classifications [8, 11, 3, 4, 2]. The passage from S^k to one coordinate uses [9, Corollary 3]. The involution input for $\text{PSp}_4(2^f)$ is

[10, Corollary 2.3]. No property of a section is inferred from CMP; only quotient closure is used.

The July 2026 Kourovka Notebook still prints Problem 18.68 without a solution annotation. That dated bibliographic fact is not needed in the proof and is not a claim that no unindexed or contemporaneous solution exists.

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